

LAND USE AND AVIAN SPECIES DIVERSITY ALONG AN URBAN GRADIENT¹

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Abstract. I examined the distribution and abundance of bird species across an urban gradient, and concomitant changes in community structure, by censusing summer resident bird populations at six sites in Santa Clara County, California (all former oak woodlands). These sites represented a gradient of urban land use that ranged from relatively undisturbed to highly developed, and included a biological preserve, recreational area, golf course, residential neighborhood, office park, and business district.

The composition of the bird community shifted from predominantly native species in the undisturbed area to invasive and exotic species in the business district. Species richness, Shannon diversity, and bird biomass peaked at moderately disturbed sites. One or more species reached maximal densities in each of the sites, and some species were restricted to a given site. The predevelopment bird species (assumed to be those found at the most undisturbed site) dropped out gradually as the sites became more urban. These patterns were significantly related to shifts in habitat structure that occurred along the gradient, as determined by canonical correspondence analysis (CCA) using the environmental variables of percent land covered by pavement, buildings, lawn, grasslands, and trees or shrubs. I compared each formal site to four additional sites with similar levels of development within a two-county area to verify that the bird communities at the formal study sites were representative of their land use category.

Key words: *birds; canonical correspondence analysis; conservation; development; gradient analysis; human disturbance; land use; species diversity; species richness; urbanization.*

INTRODUCTION

Land development and urbanization present particular challenges to conservation biologists. Although urban expansion has resulted in the conversion of cropland, pastures, and forests into built environments on a massive scale (Alig and Healy 1987), little is known about the effects of urbanization on ecosystems, communities, species, and populations because ecologists have traditionally worked in pristine, or relatively pristine, environments (Cairns 1988).

The objective of this study was to wed one well developed paradigm for ecological research, gradient analysis (*sensu* Whittaker 1967), with a new area of research for ecologists and conservation biologists, the effects of urbanization (Matson 1990). Specifically, I examined the responses of bird species and communities to urbanization by estimating summer resident bird densities in a series of habitats graded from relatively pristine to highly urban, and by comparing these estimates to human-induced changes in the landscape. The sites, all historically oak woodland, included a biological preserve, an open-space recreational area, a golf course, a residential area, an office park, and a downtown business district.

Since Whittaker (1967) championed the idea of examining plant communities with gradient analysis, the technique has been applied to studies of avian, mammalian, and aquatic communities, as well as to research on the functioning of ecosystems (for a concise review, see ter Braak and Prentice [1988]). Only recently, though, have ecologists considered employing the technique to examine the complex factors that occur in urban areas (McDonnell et al. 1993). Urban gradient analysis is not a simple examination of a feature in linear space, but rather a complex examination of the many, layered, interacting factors that reflect the fragmented and patchy nature of urban land use. Additionally, some factors may change monotonically from urban to rural areas (e.g., temperature or area covered by concrete), whereas others may peak at intermediate points along the gradient (e.g., amount of irrigation).

The effect of urbanization on bird communities has been a limited area of research, and has been only marginally considered over a full gradient of urbanization. Most researchers have compared the pre- and postdevelopment bird communities at a site (Graber and Graber 1963, Batten 1972, Walcott 1974, Aldrich and Coffin 1980), or have simultaneously compared two sites of differing levels of development (Tomialojc 1970, Emlen 1974, DeGraaf and Wentworth 1981, Beissinger and Osborne 1982, Bezzel 1985, Rosenberg et al. 1987). A few have attempted to examine a range of development by comparing residential areas of dif-

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ferent ages (Guthrie 1974, Vale and Vale 1976, Jones 1981) or a few types of land use (Hohtola 1978, Lancaster and Rees 1979, Green 1984, Ruzsczyk et al. 1987, Jokimaki and Suhonen 1993). These researchers have identified several general patterns. First, all found that species composition changes in an area as it becomes urbanized. Second, all but one confirm that the number of species decreases with increasing urbanization. Last, all agree that bird density or abundance increases with urbanization.

Examination of an urban gradient suggests that the abundance and type of resources upon which birds depend, such as food, water, perches, roosts, and nesting sites, should change greatly with development (Emlen 1974, Mills et al. 1989). Typically, moderate development will increase ornamental vegetation, water sources, primary productivity, and the amount of edge between habitats. Extreme development, however, may decrease all of these factors when they are permanently replaced with pavement and structures (Whitney and Adams 1980, Godron and Forman 1983, Mooney and Gulmon 1983, Rudnicki and McDonnell 1989, Sisk 1991).

Because they affect resources that are linked to bird abundance and distribution, these urbanization changes should be reflected by individual bird species and the bird community as a whole. Some species, which I call "urban exploiters," should be adept at exploiting these changes and, consequently, should reach their highest densities in developed sites. Others, "urban avoiders," should be particularly sensitive to human-induced changes in the landscape and consequently, should reach their highest densities at the most natural sites. Finally, some "suburban adaptable" species should be able to exploit the additional resources, such as ornamental vegetation (Emlen 1974, Beissinger and Osborne 1982, Rudnicki and McDonnell 1989), that may accompany moderate levels of development. These changes in resource availability may alter community patterns as well.

Another way to assess the overall changes induced by urbanization is to focus on the predevelopment bird community. Species found in the native bird community should drop out across the urbanization gradient as the amount of native habitat is reduced and then eliminated. This does not imply that a species from the original community may not be suburban adaptive, and actually increase in density with some degree of development (cf. Mills et al. 1989). For this study, I assume that the bird community at a biological preserve is the best representative of the original community.

In short, with the help of gradient analysis, I examine whether or not: (1) urbanization affects the distribution and abundance of birds; (2) individual bird species respond differentially to increasing urbanization; (3) urbanization affects bird species richness, diversity, and density; (4) urbanization adversely affects the native bird community; and (5) these patterns relate to

changes in habitat structure, which serves as a measure of resource availability.

METHODS

Field sites

The sites for this study are located within a 3-km radius centered at Stanford University near Palo Alto, California. I selected the sites as representative of typical forms of development in an urban-suburban matrix (Clay 1994). All sites have been developed since the 1890s, when the University and Palo Alto were founded, and all but the business district are located on university-owned land. I selected all sites to be as similar to one another as possible prior to development, based on the presence of mature coast live oak (*Quercus agrifolia*), valley oak (*Q. lobata*), or blue oak (*Q. douglasii*), and on early accounts of vegetation in the Palo Alto area (Cooper 1926).

The six sites include (1) biological preserve: the Jasper Ridge Biological Preserve is a 512-ha property traditionally used for biological research. Prior to 1974, portions of the preserve were leased for cattle grazing and public recreation. Currently, access is limited to researchers and visitors on docent-led tours; (2) open-space recreation area: this area was heavily grazed until 1989 and currently is used by joggers, dog-walkers, hikers, and equestrians. Motorized access is limited to maintenance vehicles; (3) golf course: the Stanford University Golf Course borders the open-space recreation area and was originally developed in 1930; (4) residential area: this area consists of single-family detached houses built mostly between 1930 and 1960. The larger area dedicated to housing (within which the study site is contained) shares borders with the open-space recreation area and the office park; (5) office park: the Stanford Research Park was created in the mid-1950s. The development of the specific study site began in 1965 and consists primarily of low-rise office buildings, parking lots, and landscaped areas. The research park borders the residential area; and (6) business district: downtown Palo Alto is separated from Stanford University buildings by an ≈ 1 km wide band of eucalyptus and oak woodland.

Survey methods

I estimated bird densities using variable circular plots (Reynolds et al. 1980). I established 16 survey points, each ≥ 100 m from its nearest neighbor, within each site. A field assistant and I surveyed each site a total of eight times in June and July 1992 and four times in June 1993 for a total of 192 point visits within each site. The open-space site was only visited six times in 1992 because it burned on 10 July 1992. A typical daily site survey began at dawn, consisted of 5-min visits to each of the 16 points, and lasted ≈ 2 h. We recorded the species of all birds identifiable by sight or song, and the distance from the point to each bird

in 5 m wide increments up to 30 m from the point, and in the two 10 m wide increments between 30 and 50 m from the point. We did not record high-flying individuals, such as vultures and crows, that passed over the circle but did not land or forage within its perimeter.

I estimated the type and amount of land cover at each site from recent aerial photographs provided by Stanford University, the City of Palo Alto, and Jasper Ridge Biological Preserve. I calculated the area covered by buildings, pavement, lawn, grassland, and trees or shrubs in a 36.6- to 48.8-m radius (depending on map scale) centered at each survey point, and then converted the total amount to percentage of site covered.

Surveys of human activity were conducted simultaneously at all of the sites except the biological preserve (where there was no activity) on the morning of 15 April 1993 between 0800 and 1100. Two-person survey teams counted the number of pedestrians, bicyclists, and cars passing within 22 m (0.16-ha circles) of a point for 15 min. Six to eight points were surveyed within each site. Survey teams also noted the number of cars parked within each of the 0.16-ha circles.

Data analysis

I determined the rank order of the sites from most natural to most urban using the Delphi technique (Zuboy 1981, Blair 1994). Faculty, staff, and students ($n = 14$) working with the Center for Conservation Biology at Stanford were asked to rank the sites independently and anonymously from most natural to most urban, and to give the criteria they used. The results of the ranking were then distributed to the participants and the process was repeated until the group reached relative consensus on the ordering.

I calculated the density of each bird species based on the technique of variable circular plots, which allows for adjustments in detectability for individual species and sites (Reynolds et al. 1980). Using all occurrence data, I created histograms of the density of each species in each distance band for each site. I visually examined these histograms to locate the band in which a species' density (and, thus, detectability) was highest, and then excluded from further analysis at that site all records of that species occurring in bands at a greater distance from the survey point than the peak band. For example, in the office park, the histogram for Scrub Jay occurrence had no discernible peak, implying that jays were detected at the same rate in all of the bands up to 50 m. Consequently, I used all records of Scrub Jay occurrence and calculated densities based upon a 50-m radius. In contrast, Bushtits peaked at 25 m, implying that not all Bushtits were detected beyond that distance. Consequently, I excluded all records of occurrence at distances >25 m, and calculated densities based upon a 25-m radius.

Using these detectability adjustments, I calculated densities by combining point counts within a site per

survey day, and then calculated the mean daily density of each species at each site. I combined the 16 point counts into a single daily count in order to meet assumptions of normality. I restricted the analysis to those species that were $>1\%$ of all observations within a site to assure that only resident species were considered; most of the excluded species were seen only one or two times within a site. At the golf course, I lowered this criterion to 0.5% of observations to accommodate the numerous transient bird flocks (e.g., Red-winged Blackbirds) that numerically overshadowed some of the resident, although less common, species.

I calculated species diversity using the Shannon Index (Shannon and Weaver 1963, Magurran 1988). I defined species richness for each site as the total number of species recorded, because effort and area covered were approximately equal at all sites. The average number of birds per hectare was the sum of the estimated daily density of all species. The grams of birds per hectare was the sum, over all species, of the estimated daily densities of each species multiplied by the average body mass of that species, according to Dunning (1993).

I used canonical correspondence analysis (CCA), performed with the program CANOCO (ter Braak 1992), to determine the influence of environmental factors on the distribution of individual species. CCA is a constrained ordination technique that incorporates the unimodal response of species to environmental variables. Linear combinations of environmental variables are selected to produce maximum separation of the species' distribution in the ordination space (ter Braak and Prentice 1988, Poulin et al. 1993, Grantham and Hann 1994). In this analysis, the species data consisted of the count of individuals in each species at a point, adjusted for detectability based on the method of variable circular plots, and summed over all days. The environmental data consisted of the percentage cover of buildings, pavement, lawn, grassland, and trees or shrubs of each sampling point.

Validation of sites

I compared each formal site to four additional sites with similar levels of development within a two-county area to ascertain that the bird communities at the formal study sites were representative of their land use category. I visited, each for 1 d in July 1993, the original study sites and four different business districts, office parks, residential areas, golf courses, open-space reserves that were heavily used for recreation, and open-space reserves that were infrequently used by hikers (because there were no additional biological preserves in the area). I surveyed the bird community with the same protocol as that used at the formal sites. I then performed a cluster analysis using the Goodman-Kruskal gamma correlation coefficient on the count of individuals within each species (not densities) of all birds observed (SYSTAT 1992).

RESULTS

The rural-urban gradient

The ranking of the sites, by the Delphi technique, from most natural to most urban was: biological preserve, open-space recreational area, golf course, residential area, office park, and business district. This ranking was confirmed empirically by the preponderance of unimodal distributions in types of land cover, amounts of human use, and the distributions of individual species when the sites were ranked in this order. In other words, many of these factors were highest at one site and decreased smoothly at sites with more or less disturbance, according to this ranking (Figs. 1 and 2).

All measures of landscape cover had unimodal distributions across the urbanization gradient except the area covered by trees and shrubs, which had an additional local maximum in the residential area. The percent of land covered by trees and shrubs was highest at the preserve. The percent area of land covered by grassland was highest at the open-space recreational area. The percent area of land covered by lawn was highest at the golf course. The percent of land covered by pavement was high in both the office park and business district. The percent of land covered by buildings was highest in the business district (Fig. 1).

The intensity of human use varied among sites. The average number of pedestrians, bicycles, and moving cars passing by survey points in 15 min was greatest in the business district. The greatest numbers of parked cars was in the office park (Fig. 2).

Avian community changes along the rural-urban gradient

The densities of all bird species (Appendix) varied to some degree across the gradient (Fig. 3, Table 1). Of the 40 species encountered, 31 species had unimodal distributions and seven had disjunct (bimodal) distributions because they were absent from one or more intervening sites along the gradient. For example, the Chestnut-backed Chickadee was found in the golf course and the preserve, but not in the open-space recreational area.

Seven species were "urban avoiders," with maximum densities reached at the preserve: Dark-eyed Junco, Blue-gray Gnatcatcher, Ash-throated Flycatcher, Steller's Jay, Wrentit, Western Wood-Pewee, and Hutton's Vireo. Thirty species were "suburban adaptable": Plain Titmouse, Bewick's Wren, European Starling, Cliff Swallow, Nuttall's Woodpecker, and Violet-green Swallow (maximum densities in the open-space recreational area); Brewer's Blackbird, Scrub Jay, California Thrasher, Mallard, Pacific-slope Flycatcher, Barn Swallow, Western Bluebird, Black Phoebe, Rufous-sided Towhee, Acorn Woodpecker, Red-winged Blackbird, California Quail, and White-breasted Nuthatch (maximum densities in the golf course); Amer-

ican Robin, Brown-headed Cowbird, Mourning Dove, Anna's Hummingbird, California Towhee, House Finch, Northern Oriole, Lesser Goldfinch, and Northern Mockingbird (maximum densities in the residential area); and Chestnut-backed Chickadee and Bushtit (maximum densities in the office park). Three species were "urban exploiters," with maximum densities in the business district: Rock Dove, White-throated Swift, and House Sparrow.

Most of the original oak woodland species (defined as those found at the biological preserve) gradually dropped out along the gradient as the sites became more urbanized. The only species from the preserve that were also present in the business district were Scrub Jay, Anna's Hummingbird, and Mourning Dove (Table 2).

Most community measures were greatest at intermediate levels of development (Fig. 4). The distribution of species richness, diversity, and density were all unimodal and peaked at the golf course, open-space recreational area, and golf course, respectively. The distribution of avian biomass was bimodal, peaking in both the business district and the golf course. If Rock Doves were omitted, the distribution of avian biomass became unimodal, with a peak in the golf course.

The canonical correspondence analysis (CCA) detected a significant relation between the distribution of individual species and the environmental measures of percent area covered by buildings, pavement, lawn, grassland, and trees or shrubs at the individual sampling points. The CCA generates environmental, species, and sample plots of different scales that are then reconstructed to the same scale and superimposed upon one another. For the sake of clarity, I display the results of this CCA as two biplots: one of environmental arrows and sites (Fig. 5a) and one of environmental arrows and species (Fig. 5b).

In this study, the first two ordination axes explained 11.6% and 8.3% of the variance in the species data, with canonical eigenvalues of 0.469 and 0.334 in comparison to the sum of all unconstrained eigenvalues, which was 4.037. These percentages are a measure of the goodness-of-fit of the model, and they express the percentage variance of the weighted averages accounted for by the two-dimensional diagram (ter Braak 1986). The arrows for environmental variables in Figs. 5a and b, in conjunction with the species points, account for 49.4% of the variance in the weighted averages of the species with respect to each of the environmental variables on the first axis, and 35.1% of the variance on the second axis (ter Braak 1986, 1987).

The first two axes have species-environment correlations of 0.823 and 0.864. These correlations measure how well the extracted variation in community composition can be explained by the environmental variables and how the variation is equal to the correlation between the site scores that are weighted mean species scores and the site scores that are linear com-

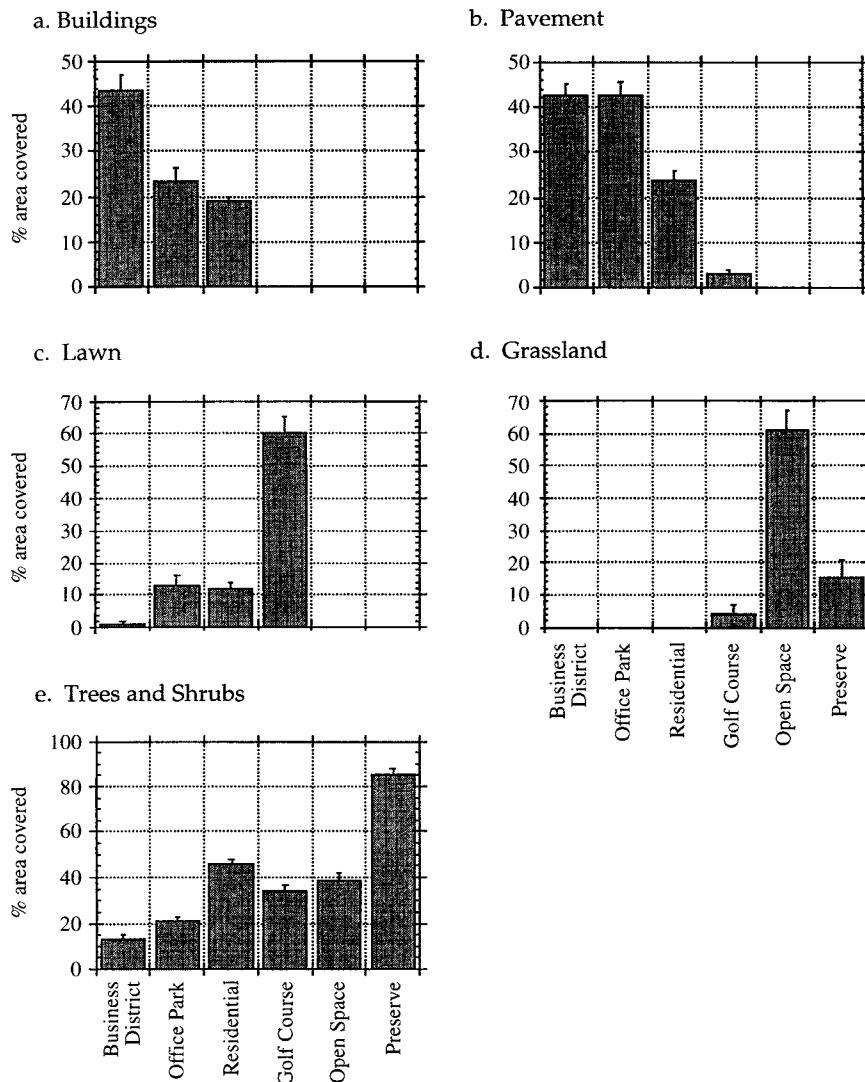


FIG. 1. Percent area (mean + 1 SE) covered by (a) buildings, (b) pavement, (c) lawn, (d) grassland, and (e) trees and shrubs as determined from aerial photographs of the six sites ($n = 16$ points within each site).

binations of the environmental variables (ter Braak 1986).

A Monte Carlo technique using 99 permutations found that the overall analysis and the two axes were significant (CCA, $P = 0.01$; Axis I, $P = 0.01$; Axis II, $P = 0.01$). Figs. 5a and b indicate that the most effective environmental variables for differentiating the sites and their bird communities were the amount of cover in buildings, lawn, and trees or shrubs. They also indicate that area covered by pavement and area covered by buildings are redundant measures.

Giving ecological meaning to the ordination axes is a heuristic, rather than a statistical, process (Liverman 1986) and, thus, not completely objective. Axis I (the horizontal axis) was successful in separating the most urbanized sites, namely the business district, office

park, and residential area. This suggests that Axis I is a measure of the built environment. Axis II (vertical) was particularly successful in separating the three least urbanized sites, namely the golf course, open-space recreation area, and biological preserve. This suggests that Axis II is a measure of vegetative complexity, with complex structure (trees and shrubs) at the top of the axis and much simpler structure (lawn) at the bottom. Interestingly, 25 of the 40 bird species are located in the top half of the diagram. If greater vegetative complexity does indeed support a greater number of species, then this reinforces the idea that Axis II is a measure of vegetative complexity.

A few simple patterns in feeding habit, size, or other life history characteristics have notable effects on the distribution of the individual species on these axes. In

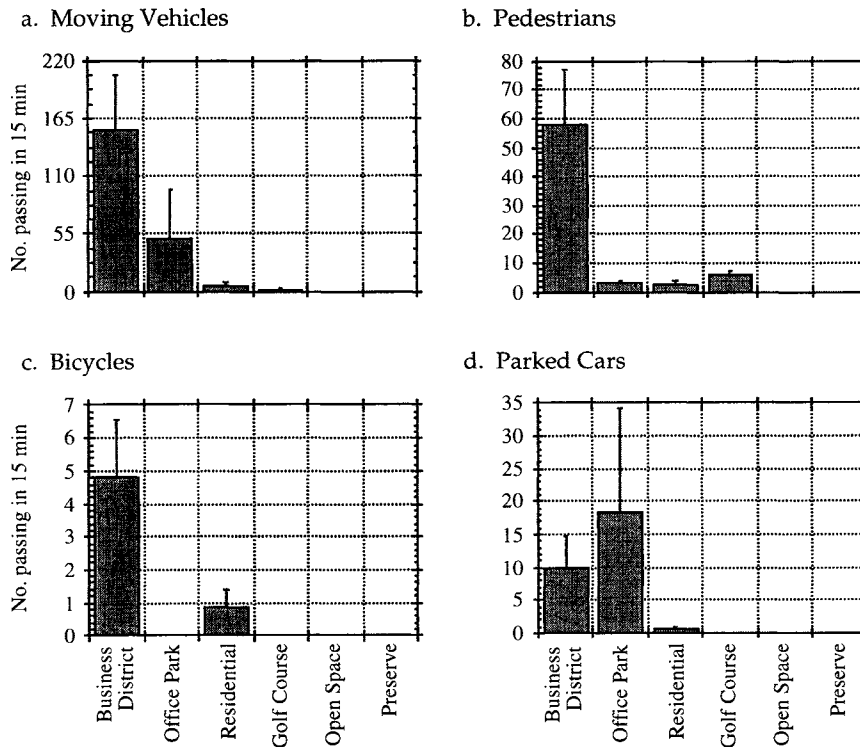


FIG. 2. Number (mean + 1 SE) of (a) moving vehicles, (b) pedestrians, (c) bicycles, and (d) parked cars passing within 22 m of a point in 15 min ($n = 6$ points within business district, office park, and residential area, $n = 7$ for golf course, $n = 8$ for open-space recreational area, $n = 0$ for biological preserve).

general, cavity nesters, such as the Plain Titmouse and the woodpeckers, are limited to the left portion of Axis I. This region represents the less-developed sites; presumably, the pattern arises because dead trees are removed from highly maintained, landscaped areas. Ground nesters, such as the quail and the junco, were limited to this area as well, presumably because less-urbanized areas support fewer predators such as house cats (Churcher and Lawton 1987). Two of the three exotics (Rock Dove and House Sparrow) were found principally in the business district, and the third (European Starling) was found in the open-space recreation area, which was the most urbanized site that also contained snags for nesting. All of these species are located on the lower portion of Axis II. Lastly, the variety of food and foraging techniques used by birds in the community tended to increase from the urban to the natural sites. For example, the ground gleaners, such as the seed-eating doves and House Sparrow, are limited to the right portion of the plot, whereas the insect foragers found in the upper left-hand quadrant use a wide variety of gleaning techniques.

Validation of sites

The bird species seen at the six formal study sites are representative of the species found at sites with similar levels of development in the San Francisco Bay area. All of the business districts, all of the office parks,

all of the golf courses, all of the preserves, four of the open-space recreational areas, and four of the residential sites were grouped by cluster analysis into distinct subsets. Additionally, office parks were grouped as a distinct subset of the residential areas. One of the residential sites and one of the open-space recreational areas (not formal study sites) grouped by themselves (Fig. 6). These results suggest that the formal study sites were more similar, in both species composition and individual species densities, to sites of similar land use than to sites of differing land use.

DISCUSSION

Urbanization affects bird diversity in two highly distinct ways. Moderate levels of development may both increase overall species diversity and decrease native bird diversity. Increasingly severe development, however, lowers both total and native species diversity. These somewhat contradictory patterns could prove misleading if only gross measures of diversity, such as species richness or Shannon diversity, are used to assess the impact of development on the bird community. Moderate levels of development apparently increase diversity (Fig. 4), but closer examination reveals that the increase in species richness results from the addition of widely distributed species at the expense of the native species (Table 2). With respect to maintaining biological diversity, it is important to differentiate be-

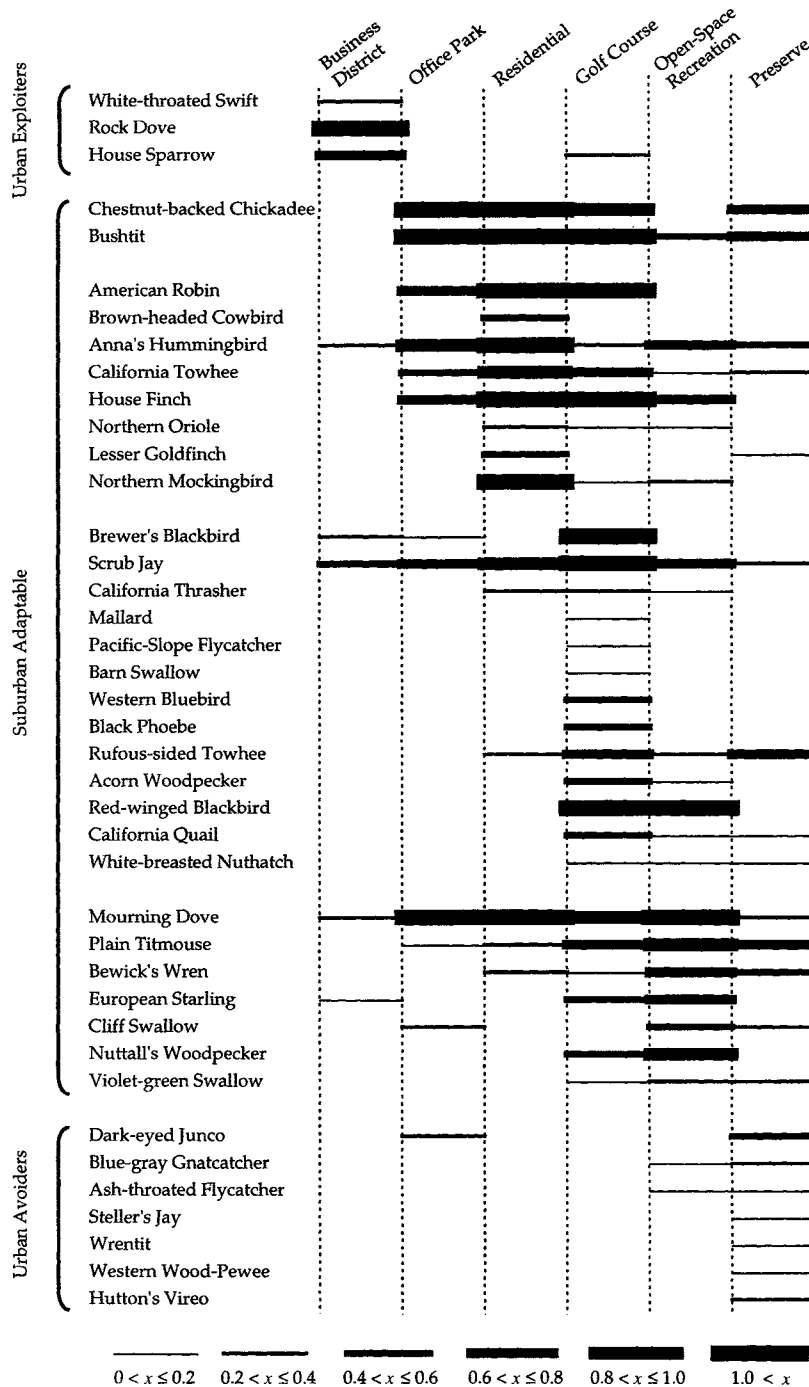


FIG. 3. Average daily densities of all bird species in all sites, arranged in order from greatest daily density in business district to greatest daily density in biological preserve (x = no. birds/ha).

tween the quality and the quantity of species that are represented in human-dominated areas.

The mechanisms that may cause these patterns should be considered. A cursory examination of the landscape changes that occur across an urbanization gradient suggests that lower levels of development, such as golf courses or residential areas with detached

housing, increase the diversity and abundance of resources available to birds. Urbanization at this level tends to alter the composition of the plant community by introducing ornamentals and changing vegetation structure (Beissinger and Osborne 1982, Rudnicki and McDonnell 1989). It may also increase structural diversity (buildings as well as vegetation) of the habitat

TABLE 1. Mean daily density (no. individuals/ha) of all bird species in all land use sites ($n = 12$ d for all sites except the open-space recreational area, where $n = 10$).†,‡

Species	Business district	Office park	Residential	Golf course	Open space	Preserve	ANOVA <i>P</i>
Mallard	0	0	0	0.11	0	0	
California Quail	0	0	0	0.51 ^a	0.15 ^b	0.05 ^b	**
Rock Dove	4.03	0	0	0	0	0	
Mourning Dove	0.41 ^a	1.19 ^{ab}	1.53 ^{ab}	0.97 ^{ab}	2.12 ^b	0.34 ^a	**
White-throated Swift	0.31	0	0	0	0	0	
Anna's Hummingbird	0.22 ^a	0.88 ^a	2.15 ^b	0.28 ^a	0.62 ^a	0.52 ^a	**
Nuttall's Woodpecker	0	0	0	0.18 ^a	0.20 ^a	0	NS
Acorn Woodpecker	0	0	0	0.45 ^a	0.20 ^b	0	**
Ash-throated Flycatcher	0	0	0	0	0.03 ^a	0.17 ^b	**
Western Wood-Pewee	0	0	0	0	0	0.09	
Black Phoebe	0	0	0	0.41	0	0	
Pacific-slope Flycatcher	0	0	0	0.09	0	0	
Violet-green Swallow	0	0	0	0.18	0.33	0.29	NS
Cliff Swallow	0	0.26	0	0	0.45	0.19	NS
Barn Swallow	0	0	0	0.11	0	0	
Scrub Jay	0.45 ^{ab}	0.70 ^a	0.84 ^a	1.79 ^c	0.83 ^a	0.28 ^b	**
Steller's Jay	0	0	0	0	0	0.05	
Wrentit	0	0	0	0	0	0.17	
Plain Titmouse	0	0.09 ^a	0.30 ^{ab}	0.68 ^b	0.81 ^b	0.66 ^b	**
Chestnut-backed Chickadee	0	1.66	1.06	0.98	0	0.62	NS
Bushtit	0	2.55	1.49	1.75	0.50	0.80	NS
White-breasted Nuthatch	0	0	0	0.18	0.18	0.09	NS
Bewick's Wren	0	0	0.32 ^{ab}	0.18 ^b	0.76 ^a	0.41 ^{ab}	**
Blue-gray Gnatcatcher	0	0	0	0	0.12 ^a	0.37 ^b	*
Western Bluebird	0	0	0	0.44	0	0	
American Robin	0	0.66 ^a	1.51 ^b	1.17 ^{ab}	0	0	**
Northern Mockingbird	0	0	1.41 ^a	0.09 ^b	0.22 ^b	0	*
California Thrasher	0	0	0.25	0.37	0.18	0	NS
European Starling	0	0.10 ^a	0	0.58 ^{ab}	0.85 ^b	0	*
Hutton's Vireo	0	0	0	0	0	0.25	
Rufous-sided Towhee	0	0	0.37 ^{ab}	0.70 ^c	0.26 ^b	0.62 ^{ac}	**
California Towhee	0	0.54 ^{ab}	0.88 ^a	0.70 ^{ab}	0.15 ^{cb}	0.22 ^{cb}	**
Dark-eyed Junco	0	0.29	0	0	0	0.48	NS
Red-winged Blackbird	0	0	0	2.89	1.40	0	NS
Brewer's Blackbird	0.37 ^a	0.05 ^a	0	1.37 ^b	0	0	**
Brown-headed Cowbird	0	0	0.55	0	0	0	
Northern Oriole	0	0	0.30	0.17	0.13	0	NS
House Sparrow	0.72 ^a	0	0	0.22 ^b	0	0	**
Lesser Goldfinch	0	0	0.58	0	0	0.15	NS
House Finch	0	0.61 ^a	3.80 ^b	1.70 ^b	0.62 ^a	0	**

* $P \leq 0.05$; ** $P \leq 0.01$; NS, not significant ($P \geq 0.05$).

† Means within a row that have the same superscript letter are not significantly different from one another (Tukey's hsd).

‡ Only sites where species were present were used in the analyses.

and provide more sites for activities such as perching (Emlen 1974, Beissinger and Osborne 1982). The introduction of watering regimes, fertilizers, and regular maintenance in urban areas may also increase primary productivity (Mooney and Gulmon 1983). In contrast, more intense development, such as office parks or business districts, could decrease the amount of resources available to birds, as construction removes substantial areas from primary production, and landscaping becomes limited to the hardiest ornamental species, which are often exotic (Whitney and Adams 1980). A similar pattern of addition and then deletion of a resource can be seen in the amount of edge habitat available in the different sites. Moderate levels of development may fragment formerly continuous habitat, increasing the amount of edge habitat available. However, edge habitat will eventually decrease with increasingly severe

urbanization, as all native habitat is eliminated with development (Godron and Forman 1983, Sisk 1991).

Results of this study suggest that individual species respond differentially to changes in resources brought about by urbanization. Three species in this study have maximal densities in the business district. Two, the Rock Dove and the House Sparrow, have been introduced by humans and are exotics in the classical sense. The third, the White-throated Swift, is native to the central coastal region of California but has expanded its range to inland Palo Alto. The swift requires cliffs for nesting but has managed to use the 'cliffs' provided by high-rise buildings in the area. These three species have been successful in co-opting resources that are evolutionarily novel to the native bird community. At the other end of the spectrum are those birds, all of which are native species, with maximal densities in the

TABLE 2. Loss of the original oak woodland bird species across the urban gradient. Only those species found at the biological preserve are listed. + symbol indicates presence.

Species	Biological preserve	Open-space recreation	Golf course	Residential area	Office park	Business district
Hutton's Vireo	+					
Western Wood-Pewee	+					
Steller's Jay	+					
Wrentit	+					
Dark-eyed Junco	+				+	
Ash-throated Flycatcher	+	+				
Blue-gray Gnatcatcher	+	+				
Cliff Swallow	+	+			+	
White-breasted Nuthatch	+	+	+			
California Quail	+	+	+			
Violet-green Swallow	+	+	+			
Rufous-sided Towhee	+	+	+	+		
Bewick's Wren	+	+	+	+		
Lesser Goldfinch	+			+		
Plain Titmouse	+	+	+	+	+	
California Towhee	+	+	+	+	+	
Bushtit	+	+	+	+	+	
Chestnut-backed Chickadee	+		+	+	+	
Scrub Jay	+	+	+	+	+	+
Anna's Hummingbird	+	+	+	+	+	+
Mourning Dove	+	+	+	+	+	+

most natural site. These species appear to decrease in number with any degree of human manipulation of the environment. In the middle of this spectrum is the most numerous group, the suburban adaptable species. These species appear to benefit from the greater quantity and variety of resources that occur with intermediate levels of development. Only two of these 30 species are non-native: the Brown-headed Cowbird (invasive) and the European Starling (exotic).

Patterns similar to these have been detected by other researchers working in urban areas. Mills et al. (1989) examined the influence of exotic and native vegetation on breeding bird densities in residential areas in Tucson, Arizona. They found that the density of territorial native birds correlated with the total vegetation volume of native plant species, whereas the density of exotic and nonterritorial birds correlated with the total vegetation volume of exotic plant species. They noted that several native bird species, including Anna's Hummingbird, Northern Mockingbird, White-winged Dove, Western Kingbird, and Hooded Oriole, occurred most frequently at urban sites. Mills et al. (1989) also described a category of birds, near natives, having densities that also correlated with total vegetation volume of exotic species. These near natives, which expanded their range into the Tucson Basin with the introduction of agriculture and urbanization, included Mallard, Inca Dove, American Robin, Great-tailed Grackle, and Bronzed Cowbird. Nine of the 10 species found by Mills et al. (1989) at urban sites, or described by them as near natives, are the same, or close relatives of, species classified as suburban adaptable in this study. These results support the idea that suburban development increases the abundance of critical resources that can be exploited by some native birds. For example,

while conducting this study, I observed Anna's Hummingbirds taking advantage of the wealth of nectar sources in the residential area, and I often found American Robins foraging on the softer ground provided by irrigation at the golf course, residential area, and office park.

One caveat should be noted: high density of a species in a site does not indicate whether that particular site is self-sustaining, a population source, or a population sink. The site actually may be relying on dispersal of individuals from other sites to maintain high densities. In fact, R. Bowman (*personal communication*) has documented such a situation with the Florida Scrub Jay. He suggests that jays can be found at high densities in suburban residential sites, but these populations are not necessarily self-sustaining.

Patterns at the community level found in this study contradict the findings of almost every other study on the effects of urbanization on birds. Contrary to what would be expected from a survey of the ornithological literature (Graber and Graber 1963, Tomialojc 1970, Batten 1972, Emlen 1974, Guthrie 1974, Walcott 1974, Vale and Vale 1976, Hohtola 1978, DeGraaf and Wentworth 1981, Jones 1981, Beissinger and Osborne 1982, Green 1984, Bezzel 1985, Rosenberg et al. 1987, Ruzsczyk et al. 1987), Shannon diversity, species number, bird density, and bird biomass (if Rock Doves were omitted) all peaked at intermediate levels of urbanization rather than at the most natural site or, with respect to biomass, at the most urban site. A potential explanation is that most previous studies examined shorter gradients of disturbance or compared only two points. Consequently, the most natural sites were more diverse than the most disturbed sites.

Aldrich and Coffin (1980), however, reached a dif-

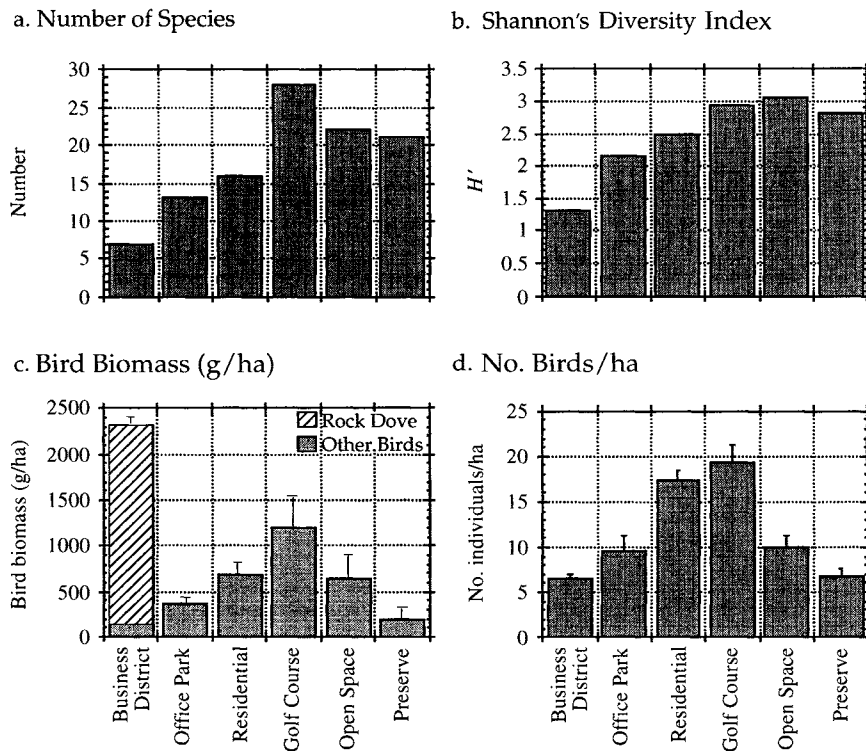


FIG. 4. Avian community measures across the urban gradient: (a) total number of bird species observed, (b) Shannon's diversity index (SE too small to display), (c) bird biomass (g/ha) with Rock Doves emphasized, and (d) number of birds per hectare. Data in (c) and (d) are based on average daily densities ± 1 SE; $n = 12$ d in each site, except $n = 10$ for open-space recreational area.

ferent conclusion than did most researchers when they compared an undisturbed oak woodland censused in 1942 to the same suburbanized site in 1979. They concluded that bird species diversity increased in urban areas. Their results are congruent with this study because the most natural area was a low diversity, undisturbed woodland and their most urbanized site was a high diversity, moderately disturbed residential area. Data from two other studies (Lancaster and Rees 1979, Jokimaki and Suhonen 1993) also indicate increases in species richness and/or diversity at slight levels of development.

These community patterns are consistent with a number of other studies, both theoretical (Connell 1978, Grime 1979, Huston 1979, Tilman 1982, Denslow 1985, Petraitis et al. 1989) and empirical (Lubchenco 1978, Paine and Levin 1981, del Moral 1983), supporting the idea that species richness peaks at intermediate levels of disturbance. This body of work relating diversity and disturbance began with the intermediate disturbance hypothesis (Connell 1978). Connell suggested that maximum diversity should be found at sites that are disturbed at intermediate spatial or temporal scales. Because disturbance disrupts those species whose superior competitive abilities generally enable them to achieve dominance in a community, disturbance permits less competitive species to co-ex-

ist. If disturbance is too frequent or too severe, only those species that are good at dispersing and/or reaching maturity quickly will remain. If disturbance is too infrequent, those species that are good competitors will dominate the community.

McDonnell et al. (1993), using slightly different reasoning than Connell's, extended the idea of natural disturbance to include human land use. They predicted that species richness should peak at intermediate levels of development because biotic limitations are high at the rural end of an urban-to-rural gradient and physical limitations are high at the urban end. This study provides the first evidence that such a pattern exists, and suggests that the pattern may be brought about by these limitations.

One question that arises from extending theories relating diversity and disturbance to include human-induced landscape disturbance is how these types of disturbance differ from natural ones. A widely used definition of natural disturbance is "any relatively discrete event in time that disrupts ecosystem, community, or population structure and changes resources, substrate availability, or the physical environment" (White and Pickett 1985). White and Pickett state that, in order to build a theoretical framework to address questions about disturbance, disturbances need to be described in terms of spatial distribution, frequency, cycle (the

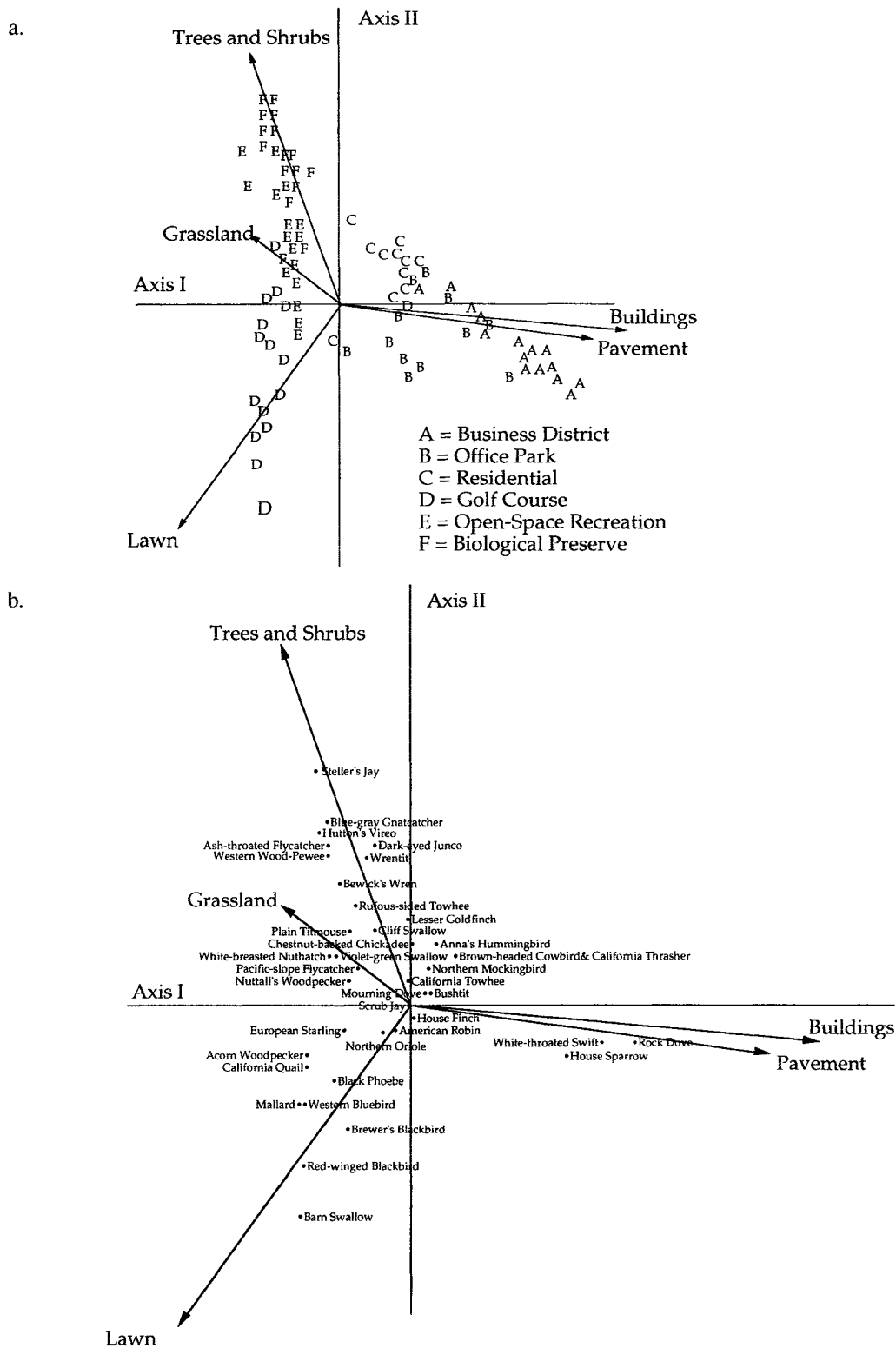


FIG. 5. Ordination diagrams of canonical correspondence analysis. (a) Sites as a linear combination of environmental measures. Letters represent points within each site ($n = 16$ for each site; some are printed on top of one another). Arrows represent direction of steepest change of environmental measures, and are oriented toward the direction of steepest increase of the variable. Arrow length indicates the importance of the environmental variable in the model, arrow direction indicates how well the environmental variable is correlated with the various axes, the angle between arrows indicates correlation

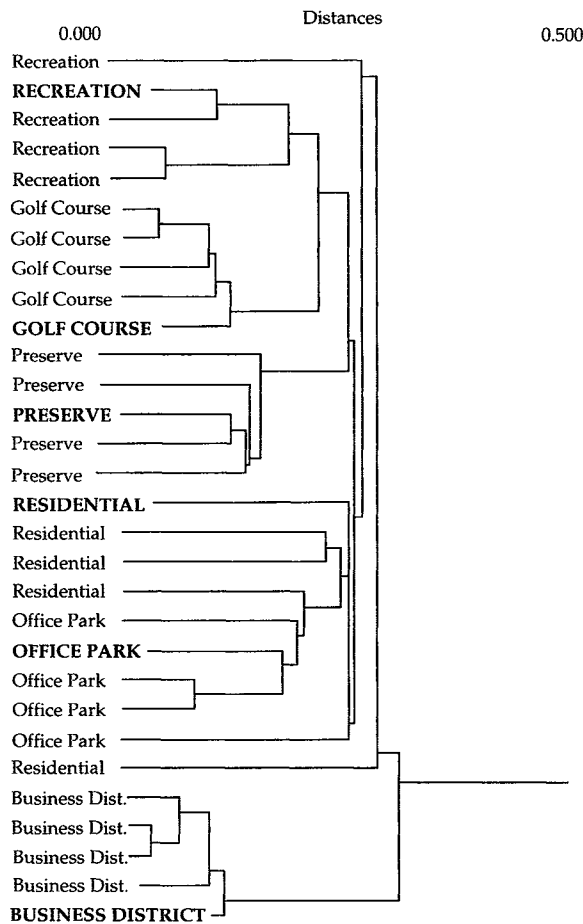


FIG. 6. Cluster analysis of validation of sites. Daily counts of species were taken at four different but comparable sites for each land use type in the Palo Alto area. These counts were clustered using distances computed with the Goodman-Kruskal gamma correlation coefficient. Formal study sites are emphasized in bold capitals.

inverse of frequency), rotation period, predictability, area, magnitude in terms of intensity (related to physical force) or severity (related to biological impact), and synergism with other disturbances.

By these standards, the disturbance caused by land development fits firmly into the framework traditionally used to describe natural disturbance, although the ways in which it differs should be highlighted. Land use development is similar to natural disturbance, such as hurricane or fire, in that it results in removal of biomass from a site, but it differs in the persistence of

that disturbance. Rarely is a developed site allowed to revert to its "natural state." Rather, it is actively maintained to prevent such reversion. In effect, groundskeepers and maintenance regimes become frequent, low-magnitude disturbances at a site after the infrequent, large-magnitude disturbance caused by its initial development. Consequently, the level of disturbance in urbanized areas should not be measured by its intensity or how often it occurs at a site, as is the case with fires or hurricanes, but rather by what percentage of the area is initially taken out of primary production and how regularly the remaining portion is maintained.

The pattern of deletion of bird species from the original oak woodland community that occurred across this urban gradient suggests that any land use development is detrimental to the native bird community. Even minor perturbations, such as the grazing that formerly occurred in the open-space recreation area, apparently lead to a loss of species. Moreover, species that disappear at the lightly disturbed sites do not reappear at some more highly disturbed site.

This finding can be used to guide land use planning on a regional scale. If planners wish to maintain predevelopment levels of biodiversity, then development should be spatially concentrated. For instance, office parks may offer a veneer of landscaping but they apparently do not provide the full array of resources required to maintain the original biological community. Consequently, in order to maximize native biodiversity, it would be advisable to concentrate business endeavors in as small an area as possible to maximize the amount of undeveloped land. This study also indicates that golf courses are not as benign a land use as some people assume (Harker et al. 1993), leading to a loss of >40% of the species found in the predevelopment community. Golf courses may be greenways with an abundance of birds and high species diversity, but they do not maintain the original species composition or abundance of the predevelopment community.

One question that arises from this study is whether or not effects of urbanization on the distribution and abundance of birds observed at the local level have a measurable effect at a larger spatial scale. Root and McDaniel (1994), using Christmas Bird Count data, analyzed state-by-state trends of 50 songbirds with winter ranges that are apparently limited by colder temperatures in the north. They concluded that species of birds that eat seeds from grasses and forbs (e.g., sparrows and meadowlarks) are declining in number in

between variables (a small angle implies high correlation), and the location of a site score relative to arrows indicates the environmental characteristics of the site (Palmer 1993, Poulin et al. 1993). The longer the arrow, the greater the correlation between the variable and the ordination axes and, hence, the greater that variable's influence in explaining the pattern of community variation. This also indicates the variable's relative predictive ability (ter Braak 1986, Grantham and Hann 1994). (b) CCA ordination of species and environmental measures. Arrows represent direction of steepest change of environmental measures. The location of a species score relative to arrows indicates the environmental preferences of that species (Palmer 1993, Poulin et al. 1993).

more states than are birds that eat seeds or berries from shrubs and trees (e.g., Tufted Titmouse and Cedar Waxwing). They suggested that this may be attributed to modification of grassland and early successional habitats to the detriment of certain species, whereas ornamental fruiting bushes, shrubs, and trees planted in urbanized areas may benefit other species. This reinforces the idea that moderate levels of development increase resource availability for some birds, and that the pattern observed in this study is also detectable at a far larger scale.

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APPENDIX

Names of all bird species mentioned in text.

Common name	Scientific name	Common name	Scientific name
Mallard	<i>Anas platyrhynchos</i>	Bushtit	<i>Psaltriparus minimus</i>
California Quail	<i>Callipepla californica</i>	White-breasted Nuthatch	<i>Sitta carolinensis</i>
Rock Dove	<i>Columba livia</i>	Bewick's Wren	<i>Thryomanes bewickii</i>
White-winged Dove	<i>Columba asiatica</i>	Blue-gray Gnatcatcher	<i>Poliophtila caerules</i>
Mourning Dove	<i>Zenaida macroura</i>	Western Bluebird	<i>Sialia mexicana</i>
Inca Dove	<i>Columbina inca</i>	American Robin	<i>Turdus migratorius</i>
White-throated Swift	<i>Aeronautes saxatalis</i>	Northern Mockingbird	<i>Mimus polyglottos</i>
Anna's Hummingbird	<i>Calypse anna</i>	California Thrasher	<i>Toxostoma redivivum</i>
Nuttall's Woodpecker	<i>Picoides nuttallii</i>	Cedar Waxwing	<i>Bombicilla cedrorum</i>
Acorn Woodpecker	<i>Melanerpes formicivorus</i>	European Starling	<i>Sturnus vulgaris</i>
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	Hutton's Vireo	<i>Vireo huttoni</i>
Western Kingbird	<i>Tyrannus verticalis</i>	Rufous-sided Towhee	<i>Pipilo erythrophthalmus</i>
Western Wood-Pewee	<i>Contopus sordidulus</i>	California Towhee	<i>Pipilo crissalis</i>
Black Phoebe	<i>Sayornis nigricans</i>	Dark-eyed Junco	<i>Junco hyemalis</i>
Pacific-slope Flycatcher	<i>Empidonax difficilis</i>	Red-winged Blackbird	<i>Agelaius phoeniceus</i>
Violet-green Swallow	<i>Tachycineta thalassina</i>	Brewer's Blackbird	<i>Euphagus cyanocephalus</i>
Cliff Swallow	<i>Hirundo pyrrhonota</i>	Brown-headed Cowbird	<i>Molothrus ater</i>
Barn Swallow	<i>Hirundo rustica</i>	Bronzed Cowbird	<i>Molothrus aeneus</i>
Scrub Jay	<i>Aphelocoma coerulescens</i>	Great-tailed Grackle	<i>Quiscalus mexicanus</i>
Steller's Jay	<i>Cyanocitta stelleri</i>	Northern Oriole	<i>Icterus galbula</i>
Wrentit	<i>Chamaea fasciata</i>	Hooded Oriole	<i>Icterus cucullatus</i>
Tufted Titmouse	<i>Parus bicolor</i>	House Sparrow	<i>Passer domesticus</i>
Plain Titmouse	<i>Parus inornatus</i>	Lesser Goldfinch	<i>Carduelis psaltria</i>
Chestnut-backed Chickadee	<i>Parus rufescens</i>	House Finch	<i>Carpodacus mexicanus</i>